

Midterm Test 2025. Row 1. Solutions

Monday, December 8, 2025

Mathematical Analysis
1st Year

Babes-Bolyai University
Bachelor in Computer Science

Midterm Test 2025. Row 1

1. Let $A \subseteq \mathbb{R}$ be a bounded set. Prove that

1p

$$\forall \varepsilon > 0, \exists x \in A \text{ such that } \sup(A) - \varepsilon < x.$$

Proof: By definition, $\sup(A) \geq x, \forall x \in A$. and if $y \geq x, \forall x \in A$, then $\sup(A) \leq y$. (1) 0.5p
Let $\varepsilon > 0$, then $\sup(A) - \varepsilon < \sup(A) \xrightarrow{(1)} \sup(A) - \varepsilon \notin \text{ub}(A) \Rightarrow \exists x \in A \text{ s.t. } \sup(A) - \varepsilon < x$. 0.5p

2. Study the convergence of the following series:

4p

(a) $\sum_{n=1}^{\infty} \frac{a^n}{a+n}, a > 0.$

(c) $\sum_{n=1}^{\infty} \sqrt{n}(\sin \frac{1}{n} - \sin \frac{1}{n+1}).$

(b) $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot (2n)}.$

(d) $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}.$

(a) $x_n = \frac{a^n}{a+n}, \frac{x_{n+1}}{x_n} = \frac{a^{n+1}}{a+n+1} \cdot \frac{a+n}{a^n} = a \cdot \frac{n+a}{n+1} \rightarrow a$. Ratio Test: if $a < 1$, convergent 0.75p
For $a=1$, $x_n = \frac{1}{n+1}, \sum_{n=1}^{\infty} \frac{1}{n+1} = \infty$ (divergent harmonic series) 0.25p if $a > 1$, divergent.

(b) $x_n = \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot (2n)}, \frac{x_{n+1}}{x_n} = \frac{2n+1}{2n+2} \rightarrow 1$ (Ratio Test is inconclusive), 0.25p

Consider $n \left(\frac{x_n}{x_{n+1}} - 1 \right) = n \left(\frac{2n+2}{2n+1} - 1 \right) = n \cdot \frac{1}{2n+1} = \frac{n}{2n+1} \rightarrow \frac{1}{2} < 1$. Raabe-Duhamel: divergent 0.25p

(c) $\sin \frac{1}{n} - \sin \frac{1}{n+1} = 2 \sin \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+1} \right) \cdot \cos \frac{1}{2} \left(\frac{1}{n} + \frac{1}{n+1} \right) = 2 \sin \frac{1}{2n(n+1)} \cdot \cos \frac{2n+1}{2n(n+1)}$. 0.25p

Recall that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, thus $\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{2n(n+1)}}{\frac{1}{2n(n+1)}} = 1 \Rightarrow \lim_{n \rightarrow \infty} n^2 \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} n^2 \cdot \frac{2}{2n(n+1)} = 1$

$\Rightarrow 1 = \lim_{n \rightarrow \infty} n^2 \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} n^{\frac{3}{2}} \cdot \sqrt{n} \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} \frac{\sqrt{n} \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right)}{\frac{1}{n^{3/2}}} = 1$ 0.5p

$\Rightarrow \sum_{n=1}^{\infty} \sqrt{n} \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right)$ has the same nature as $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$: convergent ($\frac{3}{2} > 1$) 0.25p

Method 2: By the Mean-Value Theorem $\sin \frac{1}{n} - \sin \frac{1}{n+1} = \left(\frac{1}{n} - \frac{1}{n+1} \right) \cos \frac{1}{c_n}, c_n \in \left(\frac{1}{n+1}, \frac{1}{n} \right)$.

$\Rightarrow \sqrt{n} \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \sqrt{n} \cdot \frac{1}{n(n+1)} \cos \frac{1}{c_n} = \frac{1}{\sqrt{n}(n+1)} \cos \frac{1}{c_n} \approx \frac{1}{n^{3/2}} \Rightarrow$ convergent series

(d) $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$ has the same nature as $\int_2^{\infty} \frac{1}{x\sqrt{\ln x}} dx$ 0.5p let $u = \ln x, \frac{du}{dx} = \frac{1}{x}, du = \frac{1}{x} dx$.

Then $\int_2^{\infty} \frac{1}{x\sqrt{\ln x}} dx = \int_{\ln 2}^{\infty} \frac{1}{\sqrt{u}} du = 2\sqrt{u} \Big|_{\ln 2}^{\infty} = \infty \Rightarrow$ divergent series. 0.5p

Method 2: Cauchy Condensation Test: same nature as $\sum_{n=2}^{\infty} 2^n \frac{1}{2^n \sqrt{\ln 2^n}} = \sum_{n=2}^{\infty} \frac{1}{\sqrt{n \ln 2}}$: divergent. 0.5p

3. Compute $\int_0^{\infty} e^{-\sqrt{x}} dx = I$

1p

Let $u = \sqrt{x}, x = u^2, \frac{dx}{du} = 2u, dx = 2u du \Rightarrow I = \int_0^{\infty} \frac{e^{-u}}{(-e^{-u})'} \cdot 2u du = -2ue^{-u} \Big|_0^{\infty} + \int_0^{\infty} e^{-u} \cdot 2 du$ 0.25p

$= -2 \lim_{u \rightarrow \infty} ue^{-u} + 0 - 2e^{-u} \Big|_0^{\infty} = -2e^{-\infty} + 2e^0 = 2$ 0.25p

$$= -2 \lim_{u \rightarrow \infty} u e^{-u} + 0 - 2 e^{-u} \Big|_0^{\infty} = -\underbrace{2 e^{-\infty}}_{=0} + 2 e^0 = \boxed{2} \quad 0.25p$$

4. (a) Find the Taylor series around 0 for $\sin^2 x$.

1p

(b) Find the higher order derivatives $f^{(n)}(0)$, $n \in \mathbb{N}$, for the function

0.5p

$$f(x) = \begin{cases} \frac{1-\cos x}{x^2}, & \text{if } x \neq 0 \\ 1/2, & \text{if } x = 0. \end{cases}$$

(a) $\sin^2 x = \frac{1-\cos 2x}{2}$. We use the Taylor series for $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ and obtain

$$\cos(2x) = \sum_{n=0}^{\infty} (-1)^n \frac{(2x)^{2n}}{(2n)!} = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} x^{2n}}{(2n)!} \Rightarrow \sin^2 x = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^{2n-1} x^{2n}}{(2n)!} \quad 0.5p$$

(b) Taylor series formula: $f(x) = \sum_{n=0}^{\infty} \underbrace{f^{(n)}(0)}_{=?} \frac{x^n}{n!} \Rightarrow$ to find $f^{(n)}(0)$ we can use the Taylor series.

$$\frac{1-\cos x}{x^2} = \frac{1}{x^2} \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2(n-1)}}{(2n)!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+2)!} \Rightarrow \boxed{f^{(2n+2)}(0) = 0} \text{ and}$$

$$\boxed{f^{(2n)}(0) = \frac{(-1)^n}{(2n+1)(2n+2)}} \quad 0.25p$$

5. Find the radius of convergence and the sum of the power series

1.5p

$$\sum_{n=1}^{\infty} n^2 x^n.$$

• $L = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{n^2} \right| = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^2 = 1 \Rightarrow \boxed{R=1}$ (Radius of convergence): abs conv. $\forall x \in (-1, 1)$ 0.5p

• For $x \in (-1, 1)$, $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ | Differentiate $\Rightarrow \sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2} \xrightarrow{\cdot x} \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$ 0.5p

Diff $\Rightarrow \sum_{n=1}^{\infty} n^2 x^{n-1} = \frac{(1-x)^2 + 2x(1-x)}{(1-x)^4} = \frac{\cancel{(1-x)}(1-x+2x)}{(1-x)^3} = \frac{1+x}{(1-x)^3} \Big| \cdot x \Rightarrow \boxed{\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}}$ 0.5p